Robotic Arm Simulation

3D Model Design, Assembly & Motion Simulation in SolidWorks 2022

Task 2 — CodeAlpha Robotics & Automation Internship

This report documents the complete 3D design and simulation of a multi-DOF robotic arm built using SolidWorks 2022. The arm was modeled part by part from scratch — each component individually designed, then assembled using SolidWorks mates to create a fully constrained, motion-ready robotic system. The arm includes a rotating base, two arm links, a wrist housing, servo motors at each joint, and a gear-driven robotic gripper as the end effector. A pick-and-place motion simulation was performed using the SolidWorks Motion Study module.

Parameter	Details
Software Used	SolidWorks 2022 Premium
Total Individual Parts	10 unique components
Assembly Type	Multi-body constrained assembly (SLDASM)
Degrees of Freedom	6 DOF (Base + Shoulder + Elbow + Wrist + 2x Gripper)
Joint Actuators	Servo Motors at each revolute joint
Gripper Mechanism	Gear-driven linkage gripper (2 fingers)
Simulation	Pick and Place — SolidWorks Motion Study
Output Files	Part files (.SLDPRT), Assembly (.SLDASM), Screenshots

1. Introduction

Robotic arms are one of the most fundamental and widely deployed systems in industrial automation. From automotive welding lines to surgical theatres, the robotic arm represents the core of physical automation — a machine that can reach, grasp, move, and place objects with precision and repeatability that far exceeds human capability in demanding environments.

This project involves the complete design of a multi-degree-of-freedom robotic arm using SolidWorks 2022, a professional-grade CAD and simulation platform used widely in engineering and product development. Every component of the arm was modeled individually as a SolidWorks Part file, then brought together in a SolidWorks Assembly where mates were applied to define the kinematic relationships between parts. Finally, a motion simulation was run to demonstrate a basic pick-and-place operation.

2. Degrees of Freedom (DOF)

Degrees of Freedom (DOF) in robotics refers to the number of independent movements a robotic arm can make. Each revolute joint adds one rotational DOF. A higher DOF gives the arm more flexibility to reach positions and orientations in 3D space. The arm designed in this project has 6 degrees of freedom, making it comparable in capability to standard industrial robotic arms used in manufacturing.

DOF	Joint Location	Type of Motion	Axis	Actuator
1	Base	Rotation — left/right	Z	Servo Motor
2	Shoulder	Rotation — up/down	X	Servo Motor
3	Elbow	Rotation — up/down	X	Servo + Gear
4	Wrist	Rotation — tilt	Y	Servo Motor
5	Gripper Finger 1	Linear — open/close	Y	Gear Linkage
6	Gripper Finger 2	Linear — open/close	Y	Gear Linkage
Total	**6 DOF**			**3x Servo + 1x Gear Drive**

3. Individual Part Designs

Each component of the robotic arm was designed as a separate SolidWorks Part file (.SLDPRT). The feature tree visible in each screenshot shows the sequence of operations used — Boss-Extrude for adding material, Cut-Extrude for removing material, and Fillet for smoothing edges. Below are all individual parts with their screenshots and descriptions.

Part 1: Base Bracket

The base bracket is the structural foundation of the robotic arm. It features a flat mounting plate with a curved upper profile and a central slot cutout. The curved shape and slot allow the first link of the arm to rotate and pivot. Multiple extrude and cut operations were used to create the complex profile, and fillets were applied to all sharp edges for a clean, professional finish.

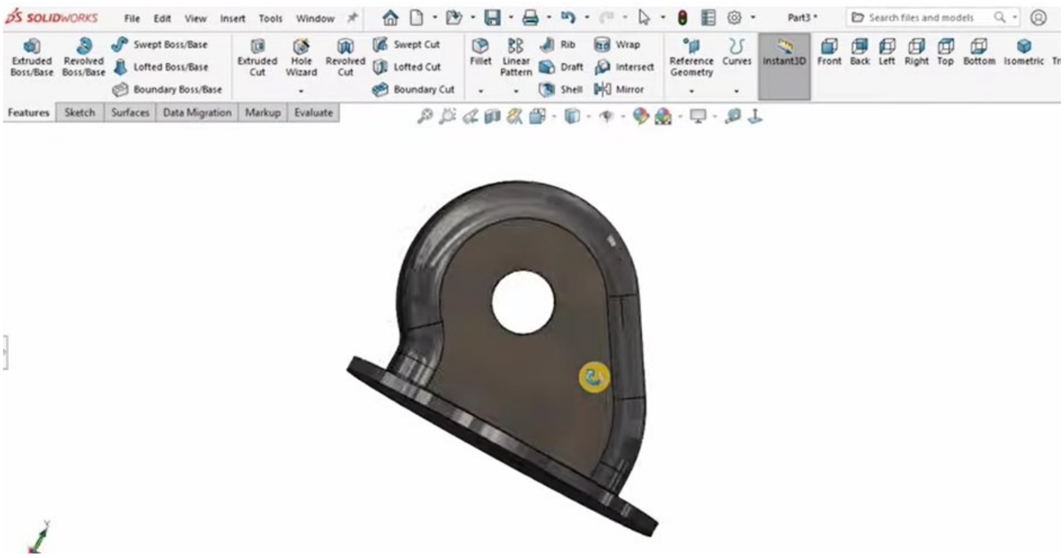

Figure 1: Part 1 — Base Bracket | SolidWorks 2022

Part 2: Base Cylinder / Rotating Ring

The rotating base is a cylindrical collar with internal ribs and cutout sections. This part sits at the very bottom of the arm and provides the primary rotation point — DOF 1. The hollowed-out interior reduces weight while the ribs maintain structural rigidity. The feature tree shows multiple Boss-Extrude, Cut-Extrude, and Fillet operations that together build up this complex shape.

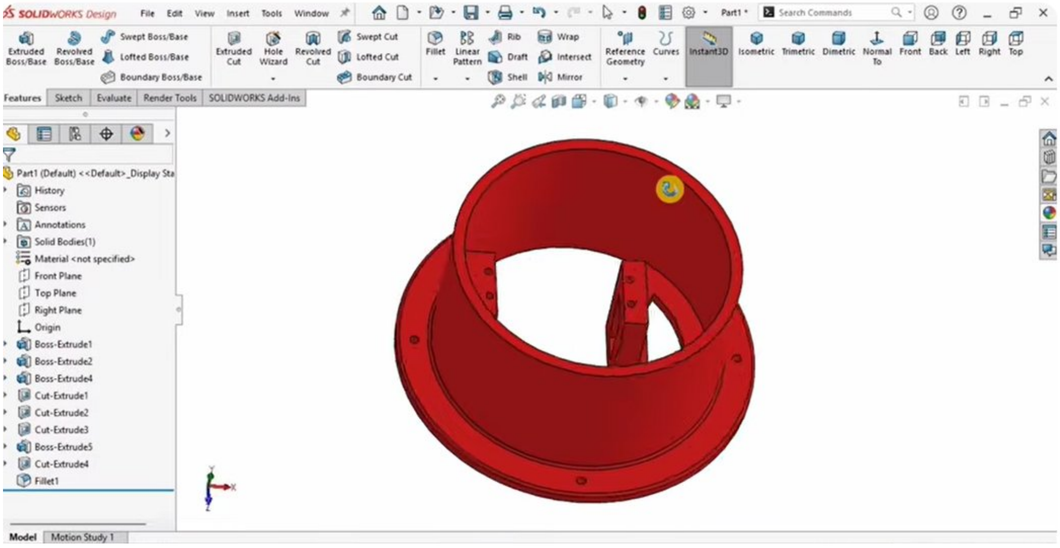

Figure 2: Part 2 — Base Cylinder / Rotating Ring | SolidWorks 2022

Part 3: Upper Arm Link

The upper arm link is an elongated connecting rod with rounded, reinforced ends. Each end has a circular boss with a hole to allow pin or bolt connections at the shoulder and elbow joints. The smooth, organic form was achieved through a main Boss-Extrude followed by two Cut-Extrude operations and multiple fillets. This part transmits force between the shoulder servo and the elbow joint.

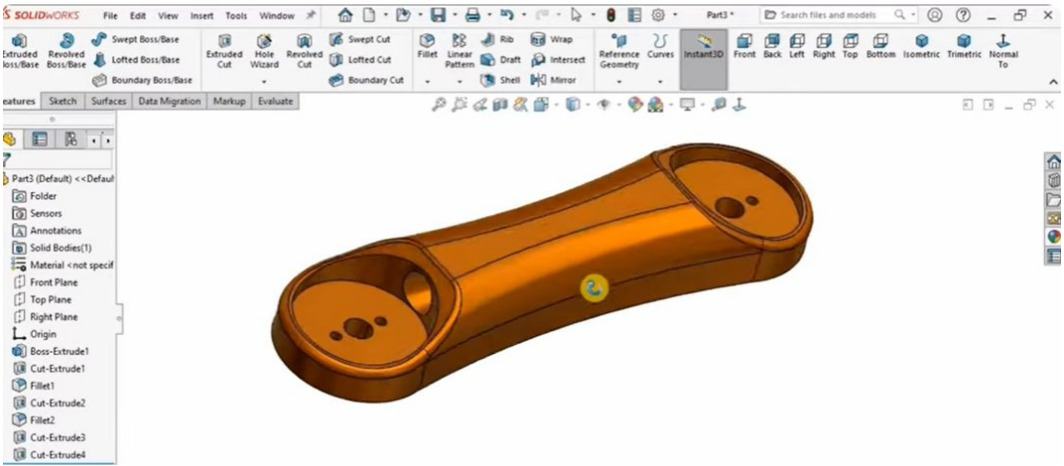

Figure 3: Part 3 — Upper Arm Link | SolidWorks 2022

Part 4: Lower Arm Body

The lower arm body is the primary structural link of the arm — the longest and most mechanically complex part. It features mounting holes at both ends for joint connections and internal pockets to reduce weight. The feature tree shows an extensive sequence of operations including multiple extrudes, cuts, and fillets, resulting in a robust, lightweight arm body suitable for carrying the upper arm and gripper assembly.

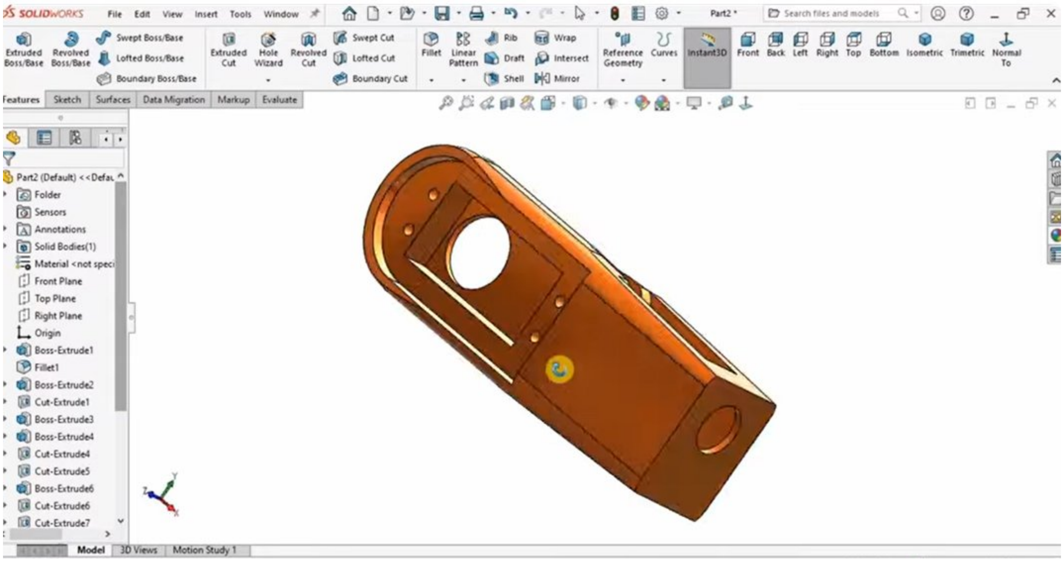

Figure 4: Part 4 — Lower Arm Body | SolidWorks 2022

Part 5: Shoulder Housing

The shoulder housing is a protective casing enclosing the servo motor at the shoulder joint. Its rounded dome shape and rectangular window cutout on the front allow the servo motor to be seated inside while keeping the assembly protected. The design uses a main extrude, two cut-extrudes for the window and internal cavity, and multiple fillets for smooth, professional edges.

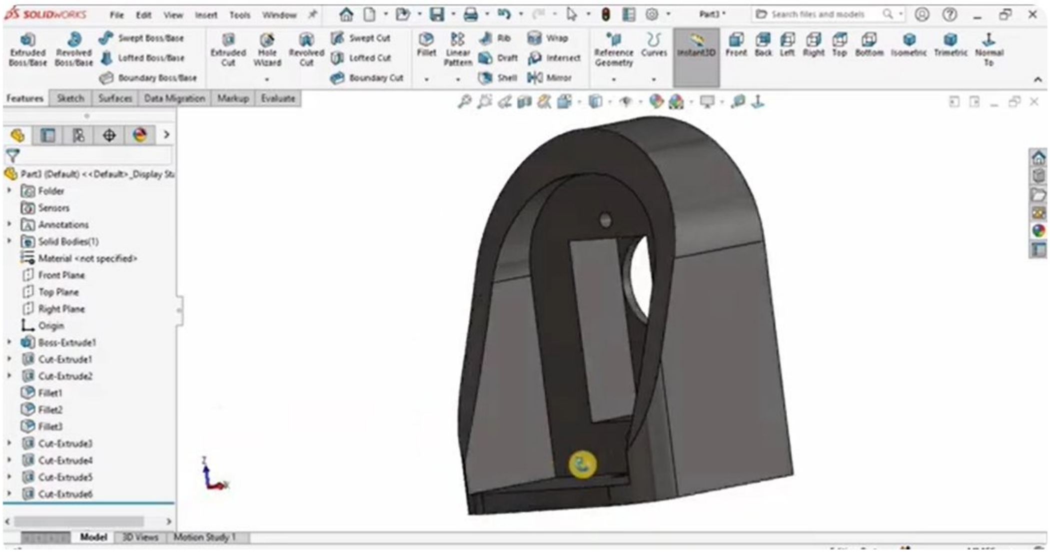

Figure 5: Part 5 — Shoulder Housing | SolidWorks 2022

Part 6: Servo Motor

The servo motor model is a detailed recreation of a standard hobby/industrial servo motor used at each rotational joint of the arm. It features the main motor body, a shaft horn on top, mounting flanges on the sides, and a chamfered edge for realistic appearance. This component is inserted multiple times in the assembly — once at the base joint, once at the shoulder, and once at the wrist — driving the arm's three primary degrees of freedom.

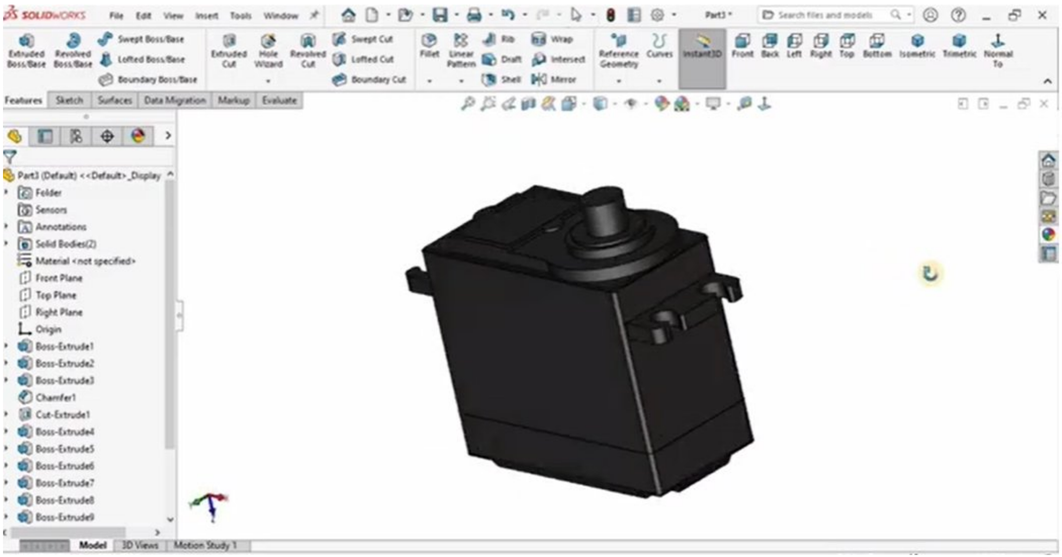

Figure 6: Part 6 — Servo Motor | SolidWorks 2022

Part 7: Gear with Arm Link

This combined part integrates a spur gear with an arm link extension. The gear teeth are formed using a single Boss-Extrude with a complex sketch, and the arm link extends from the gear hub with mounting holes at the far end. This mechanism is used at the elbow joint where the gear meshes with a mating gear to transmit rotational motion from the servo motor to the elbow link, providing mechanical advantage and controlled motion.

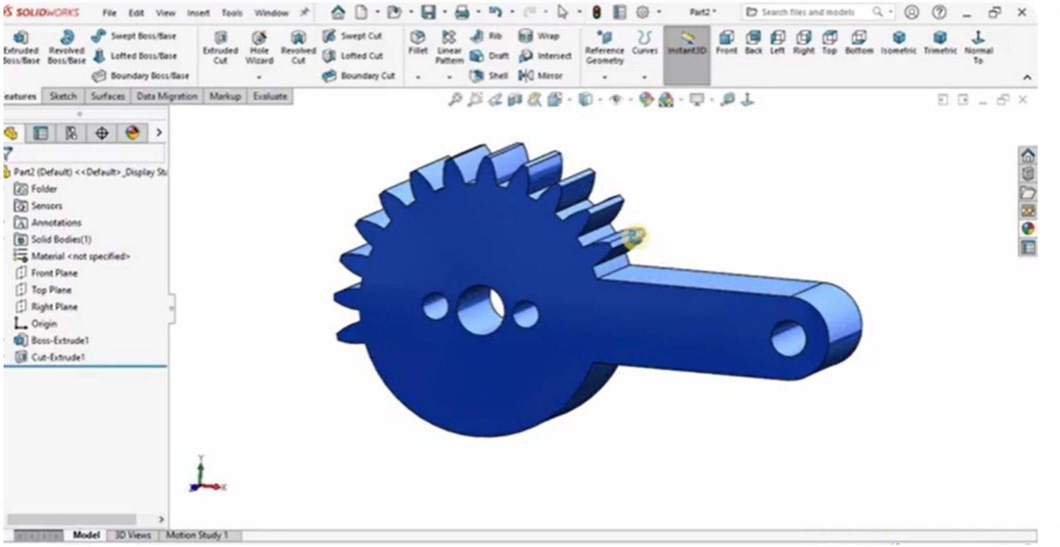

Figure 7: Part 7 — Gear with Arm Link | SolidWorks 2022

Part 8: Gripper Finger

The gripper finger is a curved, tapered link forming one half of the robotic gripper end-effector. Its angled profile allows the two fingers to wrap around objects naturally when closing. The finger has a pivot hole at the top connecting to the gripper mount and tapers to a tip at the bottom. Two of these fingers are used in the gripper sub-assembly, positioned symmetrically to open and close around target objects.

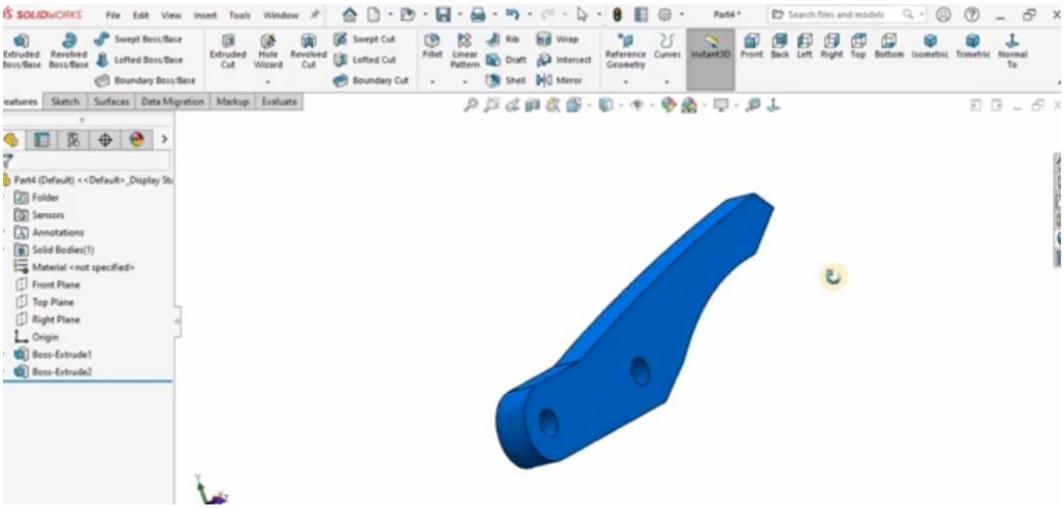

Figure 8: Part 8 — Gripper Finger | SolidWorks 2022

Part 9: Connecting Link

The connecting link is a simple but essential rigid bar with rounded, reinforced ends and circular through-holes at each end. It acts as a mechanical linkage within the gripper mechanism, converting the rotational output of the gear into the linear opening and closing motion of the gripper fingers. The clean, simple form was created with a single Boss-Extrude, making it one of the more straightforward parts in the assembly.

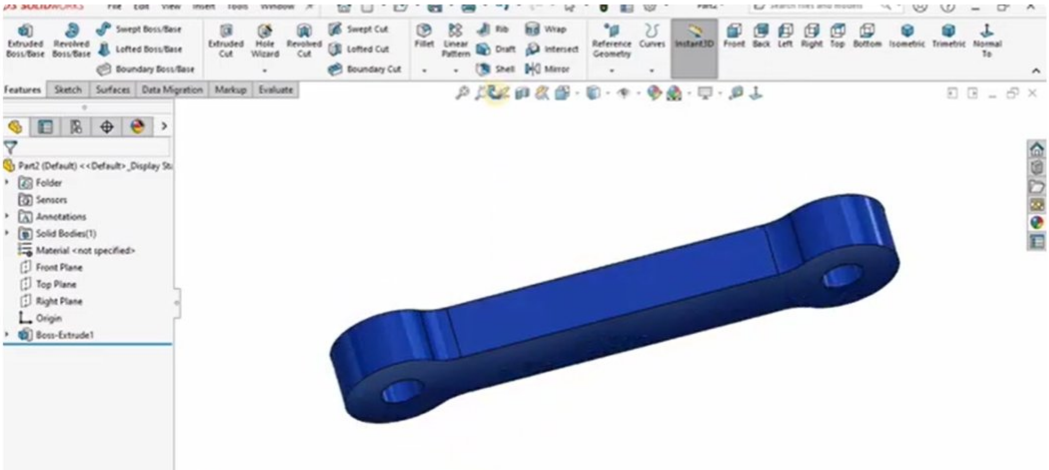

Figure 9: Part 9 — Connecting Link | SolidWorks 2022

Part 10: Wrist / Gripper Mount

The wrist mount is the most complex individual part in the assembly. It serves as the structural interface between the arm and the gripper, housing the wrist joint and providing mounting points for the gripper gear mechanism. The part features multiple extrudes, cuts, and ten individual fillets applied at various stages. The U-shaped lower section cradles the gripper assembly while the upper section connects to the final link of the arm.

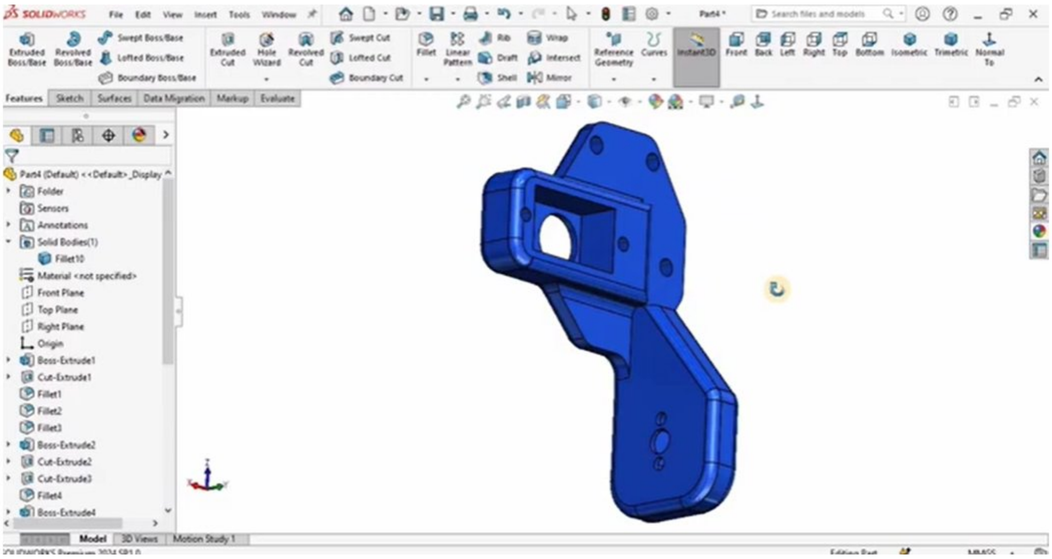

Figure 10: Part 10 — Wrist / Gripper Mount | SolidWorks 2022

4. Assembly

Sub-Assembly: Gear-Driven Gripper Mechanism

Before building the full arm assembly, the gripper was assembled as a separate sub-assembly. This approach — called bottom-up assembly design — makes the overall assembly cleaner and easier to manage. The gripper sub-assembly consists of the two gripper fingers, two connecting links, two gear components, and a servo motor. The gears mesh together so that when the servo motor rotates, both fingers open or close simultaneously in a synchronized motion. This is DOF 5 and DOF 6 of the arm.

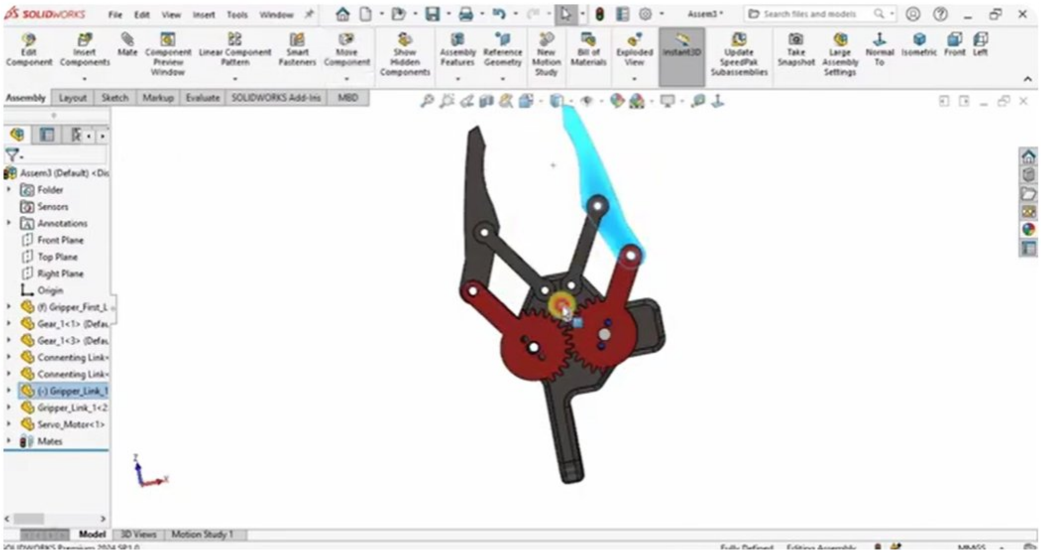

Figure 11: Gripper Sub-Assembly — gear-driven linkage mechanism with servo motor

Full Assembly: Complete 6-DOF Robotic Arm

The final assembly brings together all 10 individual parts and the gripper sub-assembly into a complete, fully constrained robotic arm. The assembly was built using SolidWorks mates: Coincident mates align faces and axes, Concentric mates align cylindrical features, and Distance mates set specific spacing between components. The result is a fully defined assembly where every part moves correctly relative to every other part.

The completed arm — visible in the final screenshot — shows all components assembled and colored by part: silver base, blue arm links, green body housing, and the multi-colored gripper assembly at the end. The arm is in an extended, ready-to-pick position with the gripper open. The feature tree on the left confirms all components are fully defined (no under-constrained parts), which is visible from the (f) and (-) symbols next to each part name.

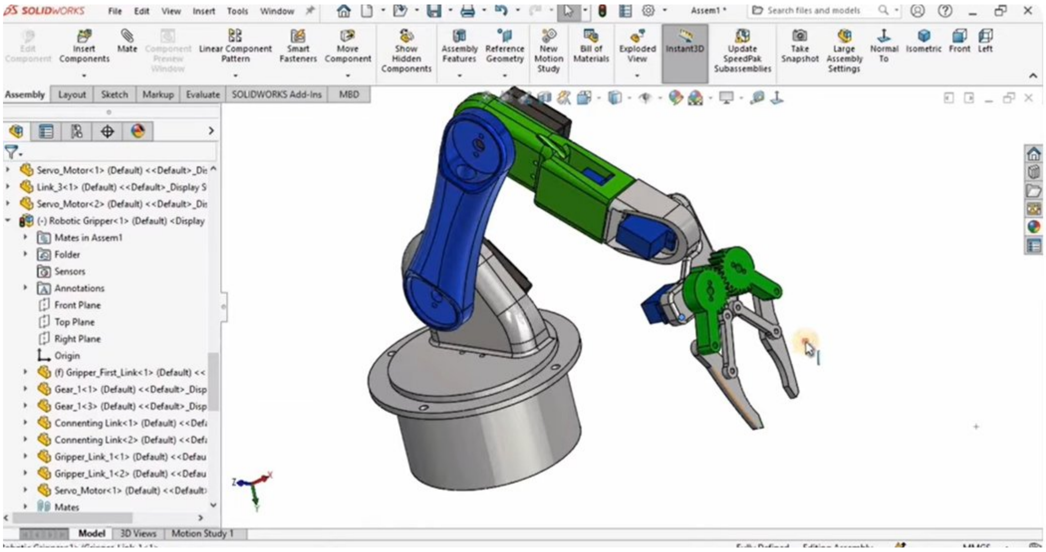

Figure 12: Full Assembly — Complete 6-DOF Robotic Arm in SolidWorks 2022

5. Pick and Place Motion Simulation

The pick-and-place operation is the most fundamental task a robotic arm performs in industrial automation. It involves reaching down to a target location, closing the gripper to grasp an object, lifting it, transporting it to a new location, and releasing it. This sequence was simulated using the SolidWorks Motion Study module, which records keyframe positions and animates the transitions between them.

Position	Time	Arm State	Gripper
Home	0 sec	Arm upright, base centred	Open
Reach	2 sec	Shoulder + elbow rotate down	Open
Pick	3 sec	Arm fully lowered to object	Closing
Lift	4 sec	Arm raises with object	Closed
Rotate	5 sec	Base rotates to target location	Closed
Place	6 sec	Arm lowers at new location	Opening
Return	7 sec	Arm returns to home position	Open

Each position was saved as a keyframe in the Motion Study timeline. SolidWorks automatically interpolates the movement between keyframes, producing a smooth animation. The servo motors at each joint were assigned rotational motion drivers, and the gear-driven gripper was given a linear driver for the open and close motion. The full animation was exported as an AVI video file for submission alongside this report.

6. Design Analysis

Workspace: With 6 degrees of freedom, the arm can reach positions and orientations throughout a large spherical workspace around the base. The combined reach of the upper and lower arm links gives an approximate working radius of 300mm to 400mm from the base centre.

Payload: The arm is designed as a prototype/simulation model. In a real implementation with the servo motors used, a payload of 200g to 500g would be appropriate depending on the servo torque rating.

Gear Mechanism: The gear-driven gripper provides a mechanical advantage that multiplies the servo motor's output torque at the gripper fingers. This is important because small servo motors need to generate enough closing force to hold objects securely.

Symmetry: The two gripper fingers are mirror images of each other, driven by meshing gears. This ensures both fingers move in perfect synchrony — one opens as the other opens, one closes as the other closes — without requiring a second actuator.

Material: In this simulation model, no material was assigned to parts. In a physical build, aluminium alloy (6061) would be suitable for the arm links for its strength-to-weight ratio, while PLA or ABS plastic could be used for the housing components.

7. Conclusion

This project successfully demonstrates the complete design workflow for a multi-DOF robotic arm in SolidWorks 2022 — from individual part modeling through assembly constraints to motion simulation. The arm features 6 degrees of freedom driven by servo motors and a gear linkage mechanism, enabling it to perform the fundamental pick-and-place task that underpins most industrial robotic applications.

The exercise reinforced several key engineering design principles: the importance of designing parts individually before assembling, the role of mates in defining kinematic relationships, and how motion simulation can validate a design before any physical prototype is built. The gear-driven gripper in particular shows how mechanical advantage can be engineered into a design to amplify force output — a concept central to both robotics and mechanical engineering.

With further development, this design could be refined with material assignments, stress analysis using SolidWorks Simulation, and more complex motion paths to simulate real production tasks. The foundations built in this project provide a solid platform for exactly that kind of advanced work.